

CENTER FOR MEDICAL INNOVATION





SUTEKI DESIGNS HSS 1-3

FREEDOM TO ENGAGE: COMFORT WITHOUT RESTRICTION

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OVERVIEW

Dropped Head Syndrome is characterized by severe weakness of the neck extensor muscles. It may occur because of neurological, neuromuscular, muscular, or other causes.

Disease states where neck extensor weakness is a problem include, but are not limited to:



<https://www.necksolutions.com/dropped-head-syndrome/>

CHALLENGES

1

Discomfort

Patients can't have their head upright in more comfortable position. It is also socially isolating.

2

No Jaw Movement, Choking Hazard

Restricted jaw movement hinders eating and talking. Head position is a choking hazard.

3

No Flexibility

Current devices offer no head movement. This increases the risk of falls while ambulatory through decreased line of sight and poor balance.



INTERVIEWS AND OBSERVATIONS

Dr. Mark Bromberg,
MD, PhD

- Knew Creighton Rider as a patient and spoke to how special he was
- Every ALS patient is so different
- Neck extensor muscle *fatigue vs weakness*
- Large benefits from allowing for jaw movement
- Works closely with Prof. Zhang in ME dept.

Dallin Sudbury,
Occupational Therapist

- Has recommended the Suteki Design device to some patients
- Challenges with constant on/off of cervical collars, especially when a caregiver is required to assist don/doffing
- DHS is a choking hazard
- Need a sitting-specific option
- Often neck extensor weakness also includes arm weakness

Cassie Kuhn,
APRN, DNP, FNP-BC

- Confirmed that there is absolutely a need, but it's a tricky one to solve
- Sitting-specific option, but also patients often just use natural chair or wheelchair headrests
- Waist bands are difficult if patient has feeding tubes
- Suteki Design is a good option for those who have difficulty swallowing – free jaw and forward head tilt

MARKET ASSESSMENT

Total Addressable Market:

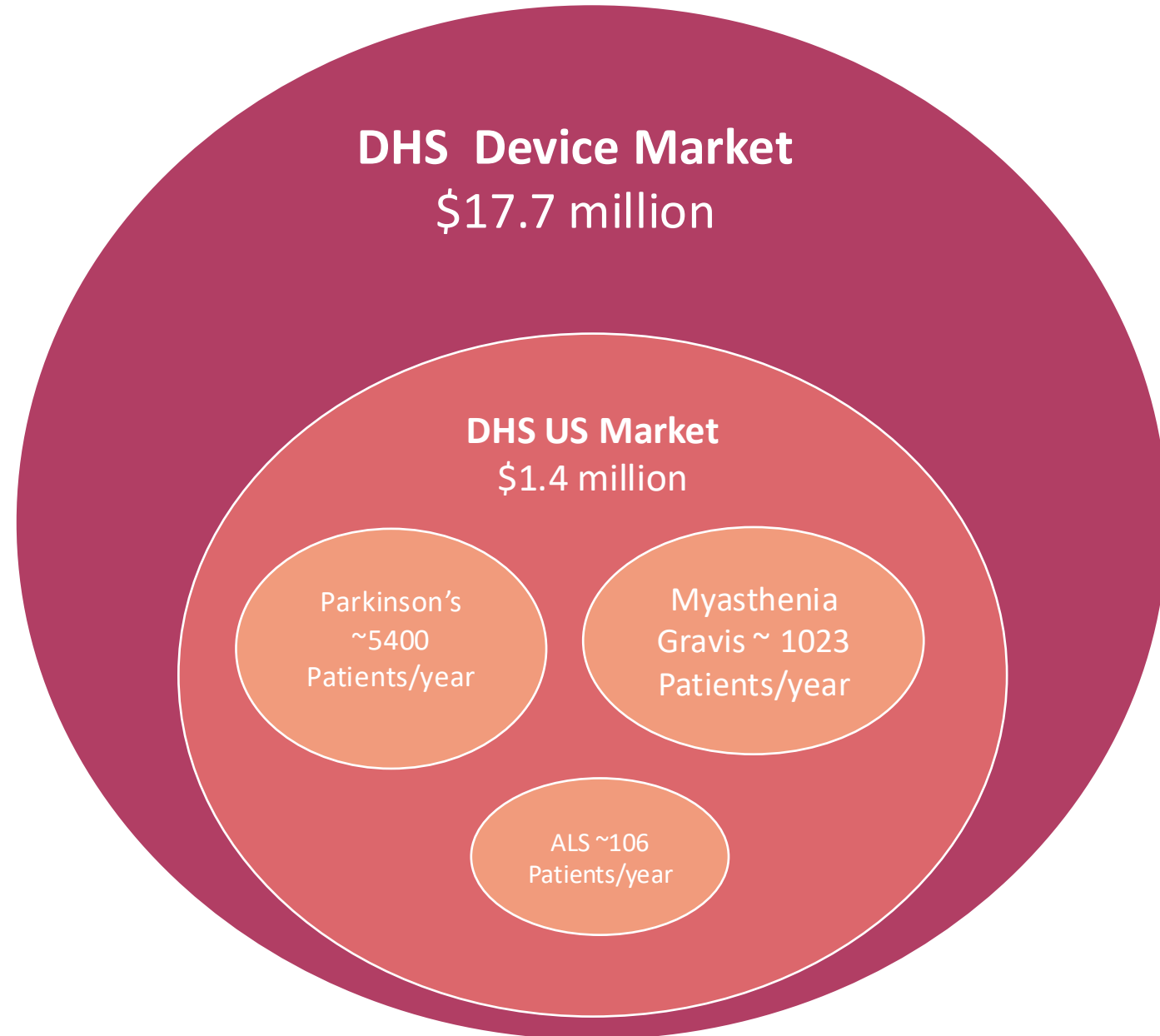
DHS Worldwide Market: **\$17.7 Million**

Serviceable Available Market:

DHS US : **\$1.4 million**

Potential Target Markets (US) per year :

- **~106 ALS** patients with DHS
- **~5400 Parkinson's** patients with DHS
- **~1023 Myasthenia Gravis** patients with DHS
- **~6529 DHS** Patients in total



Competitor Analysis

1

Dropped Head Support Band¹⁰

- Elastic Band
- Not comfortable
- Puts pressure on chest
- \$139¹⁰



2

Headmaster Collar¹¹

- Puts pressure on chin
- Not comfortable
- Can't wear for long
- \$149⁹



3

Savant Headrest¹²

- Expensive
- More used for side drop
- Needs more external help
- \$589⁸



Thoughts

The Suteki Head Support was designed with the aim of enabling individuals to live their best lives despite the challenges of DHS—a feature lacking in competitors' products.

Interview feedback highlighted that the Suteki Band offers superior mobility and comfort compared to other options available in the market.

REIMBURSEMENT CONSIDERATIONS

HCPCS ¹³			
CODE	DESCRIPTION	Floor	Ceiling
L0112	Cranial cervical Orthosis, congenital Torticollis type, with or without soft interface material, adjustable range of motion joint, custom fabricated	\$1471.15	\$1961.54
L0160	Cervical, semi-rigid, wireframe occipital/mandibular support, prefabricated, off-the-shelf (most relevant)	\$172.31	\$229.75
L0120	Cervical, flexible, non-adjustable, prefabricated, off-the-shelf (foam collar)	\$29.20	\$38.93

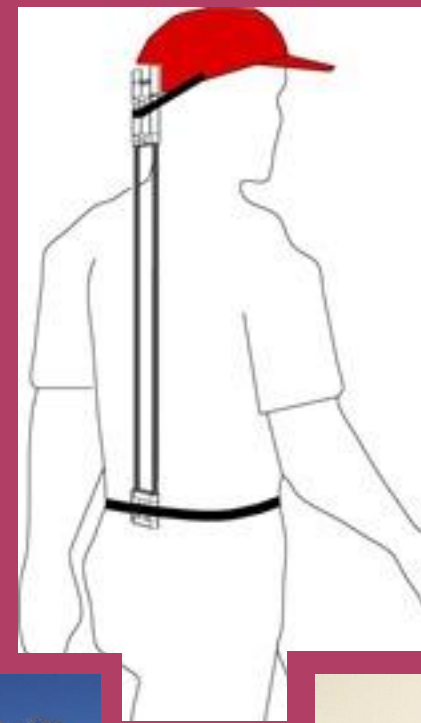
EXISTING DESIGNS

Products:

- HSS-1: Two rods down the back, attaching a hat or headband to a belt that goes around the waist
- HSS-2: Collapsible in the middle for easier transportation

Value Proposition:

- Head support
- Jaw movement
- Lateral head movement
- Elastic flexibility



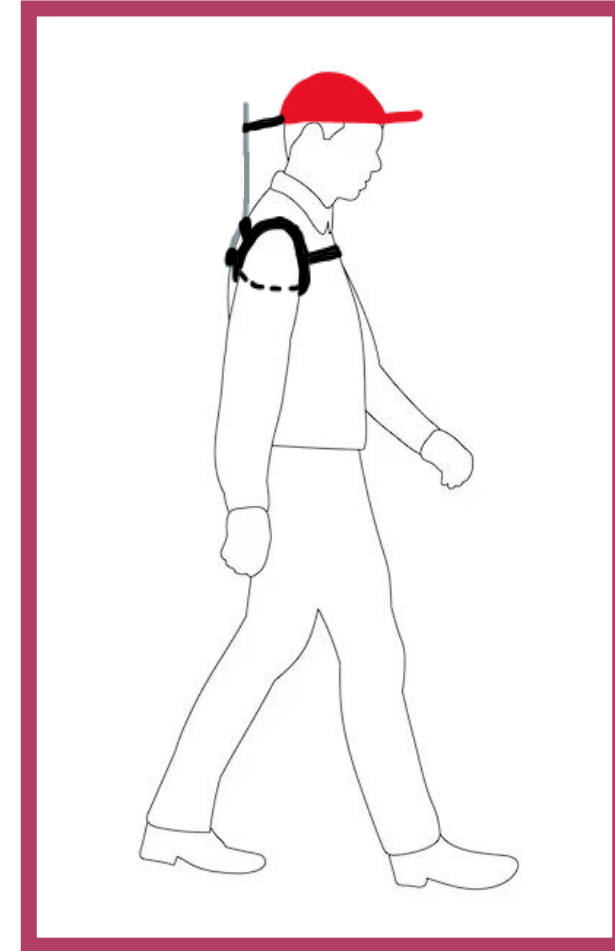
ADDITIONAL PROPOSED DEVICE

Products:

- HSS-3: To be used while sitting or standing.

Value Proposition:

- Head support
- Jaw movement
- Lateral head movement
- Elastic flexibility
- Transition from standing to sitting
- Reduced frequency of donning and doffing

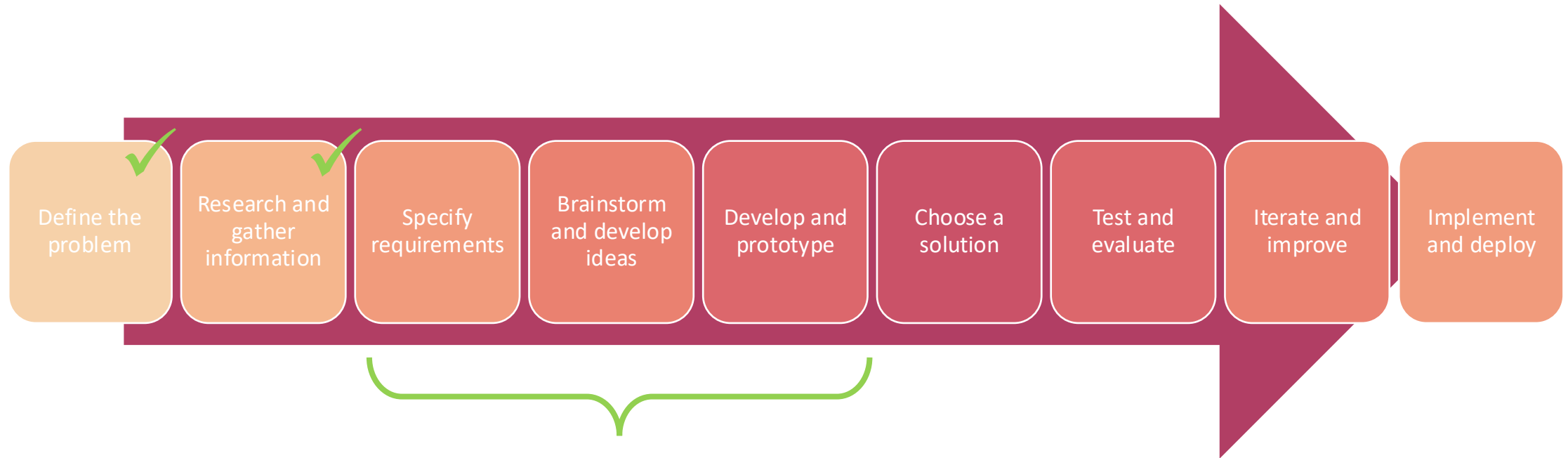


REGULATORY ASSESSMENT

CLASSIFICATION INFORMATION					
FDA CFR No.	Product Code	Class	Review Panel	Product Code Description	Submission Type
890.3490	IQK	1 – GMP Exempt	Physical Medicine	Cervical Orthosis	510(k) Exempt
890.3490	IQF	1 – GMP Exempt	Physical Medicine	Rigid Cervical-Thoracic Orthosis	510(k) Exempt

REGULATORY STRATEGIES		
Type of Submission:	☐ Traditional 510(k)	☐ Q-Sub/Pre-Submission Meeting
	☐ Special 510(k)	☐ PMA
	☐ Abbreviated 510(k)	☐ PMA Periodic Report
	☐ De Novo 510(k)	☐ PMA Supplement
	☒ Letter to File	☐ 30-Day PMA Notice
	☐ General Wellness	☐ 30-Day PMA Supplement
	☐ None	☐ 135-Day PMA Supplement
	☐ Other (IDE pre-submission)	☐ Real-Time PMA Review

NEXT STEPS



Immediate Next Steps:

- Continue to gather information, specifically:
 - User needs
 - Risk management
- Begin prototyping ideas
- Visit the ALS clinic and shadow providers
- Visit Nielsen Rehab Hospital and shadow providers



cmi.uofuhealth.org

